

ERASTEEL TECHNICAL DATASHEET: ASP 2011

Application Family: Cutting tools | Standards: Erasteel Proprietary PM High-Vanadium

1. Grade Overview & Characteristics

ASP 2011 is a high-vanadium powder metallurgical cold and knife grade boasting supreme abrasive wear resistance owing to a dense distribution of hard vanadium carbides. Non-cobalt alloy.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	2.40 - 2.56	5.0 - 5.6	1.0 - 1.4	0.0	9.2 - 9.8	0.0	High primary MC carbides

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	60.0 - 62.0 HRC	Machinability	4.0/10
Max Reachable Hardness	62.0 HRC	Wear Resistance	9.5/10
Annealed Delivery State	Max 280-350 HB	Toughness	5.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	5.0/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	4.0/10

4. Industrial Applications

Industrial knives, granulator blades, slitting cutters, paper cutting, abrasive pelletizing knives, woodworking blades.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1050-1120°C. Double tempering at 540-560°C. Typical working hardness 60-62 HRC.